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(54) SYSTEM AND METHOD FOR ELECTRIC MACHINE TORQUE ESTIMATION

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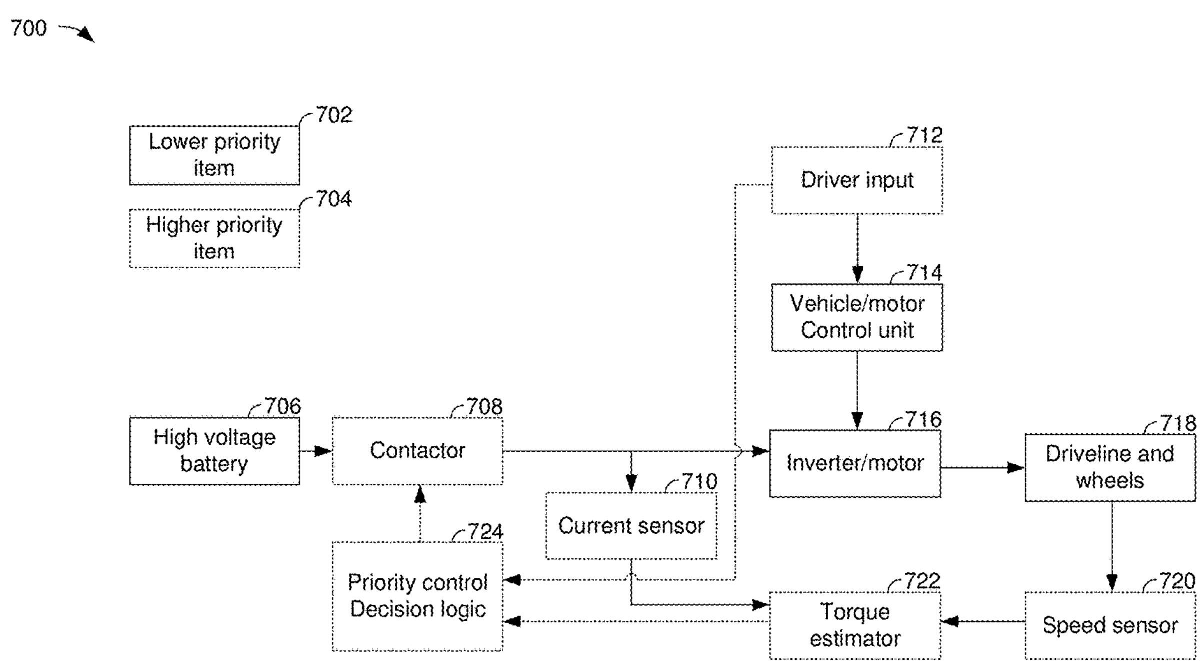
(57) **ABSTRACT**

Methods and systems for estimating torque of an electric machine that propels a vehicle are described. The methods and systems may generate torque estimates that may be applied to higher and lower priority systems. In one example, a contactor may be selectively opened according to a torque estimate that is based on interpolating between two curves.



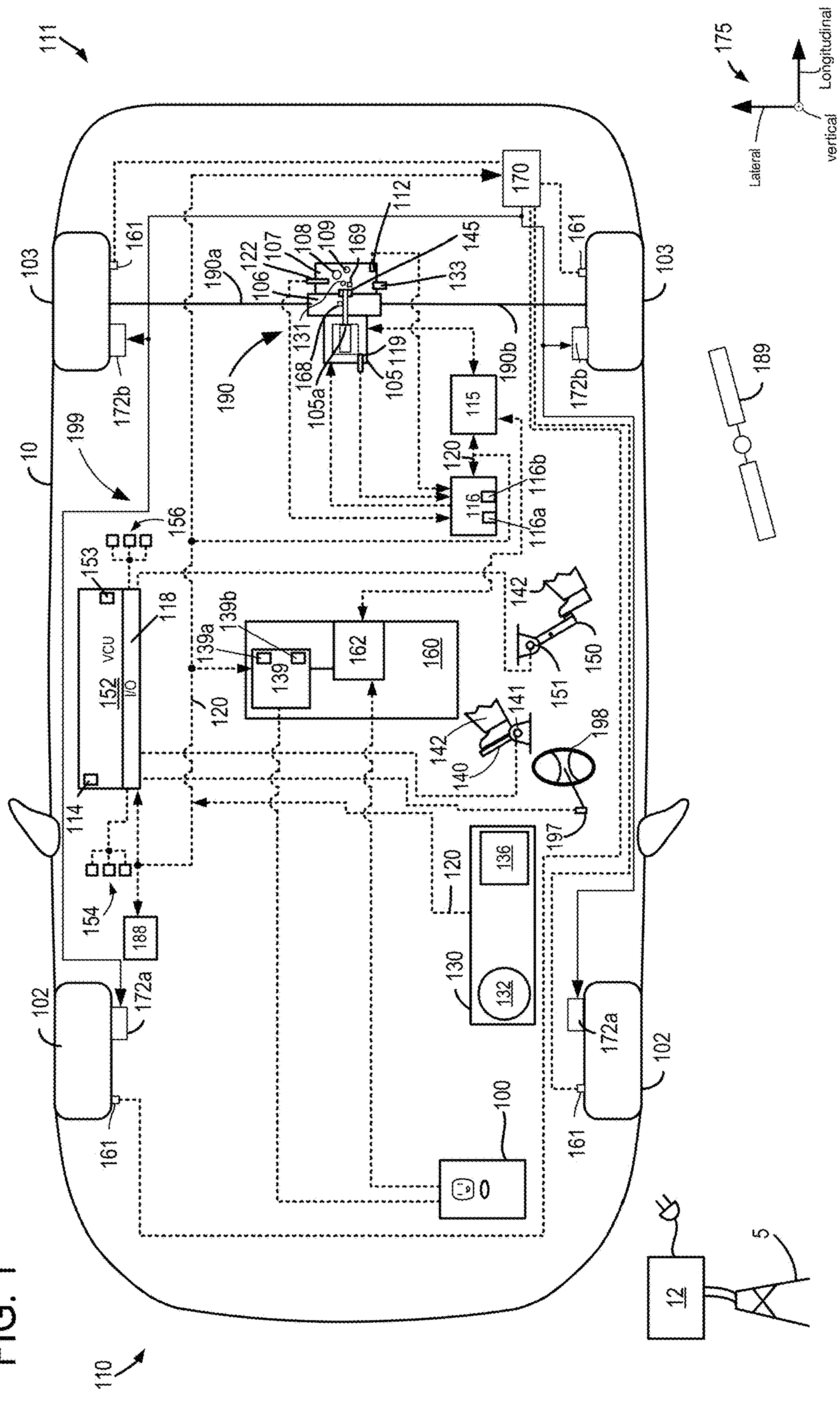


FIG. 1

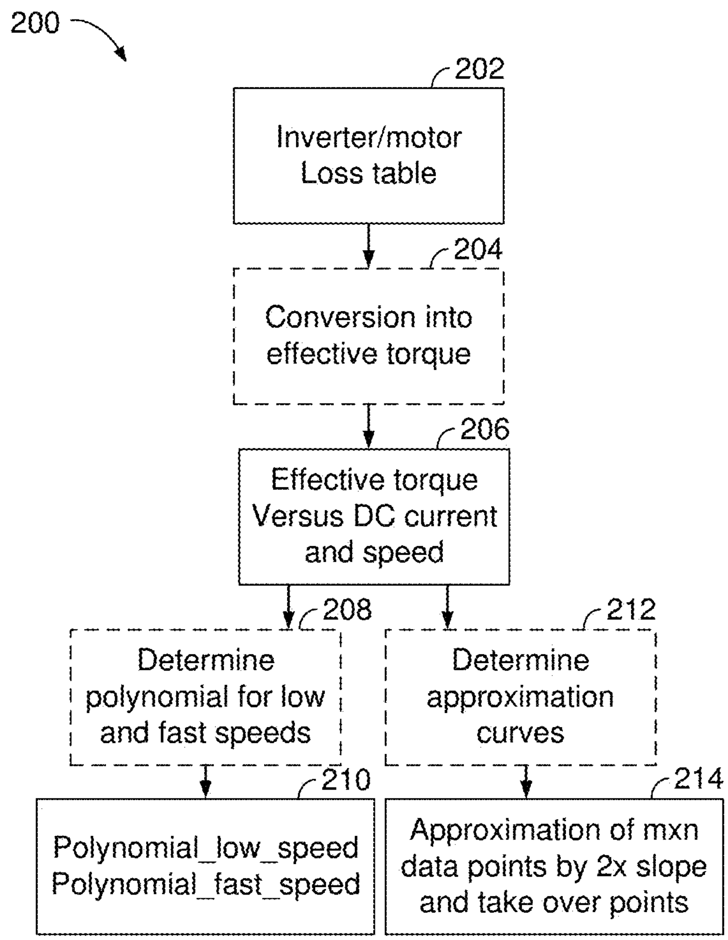

200
202
Inverter/motor
Loss table
204
Conversion into
effective torque
206
Effective torque
Versus DC current
and speed
208
Determine
polynomial for low
and fast speeds
212
Determine
approximation
curves
210
Polynomial_low_speed
Polynomial_fast_speed
214
Approximation of mxn
data points by 2x slope
and take over points

FIG. 2

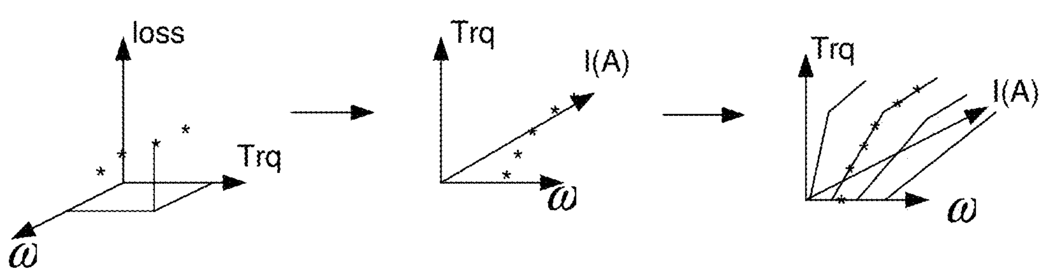

loss
Trq
ω
Trq
I(A)
ω
Trq
I(A)
ω

FIG. 3

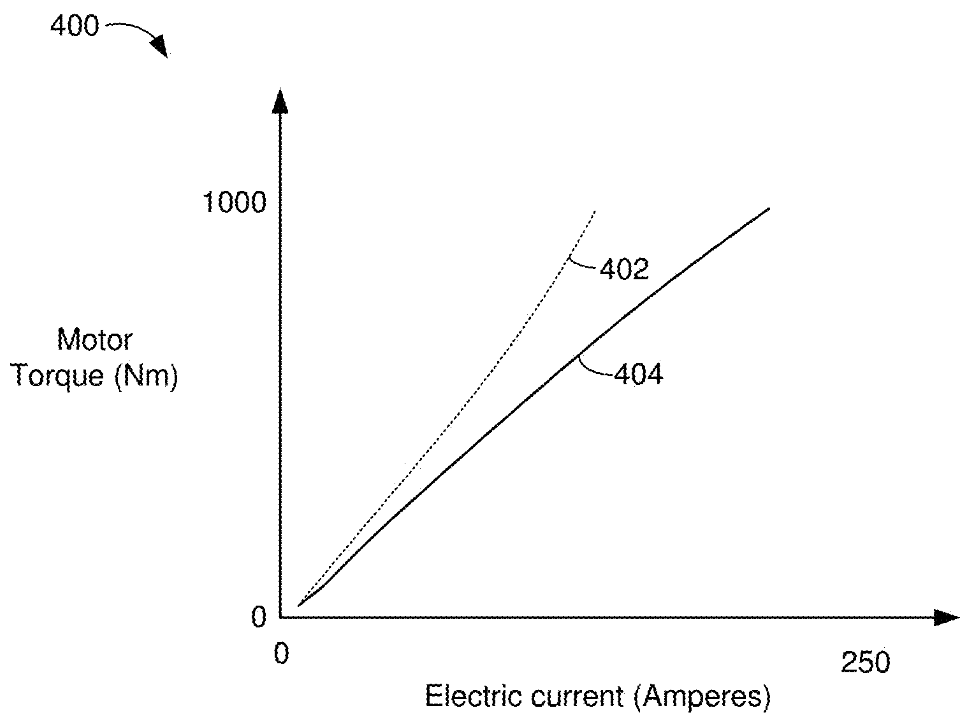

400
1000
402
Motor
Torque (Nm)
404
0
0
250
Electric current (Amperes)

FIG. 4

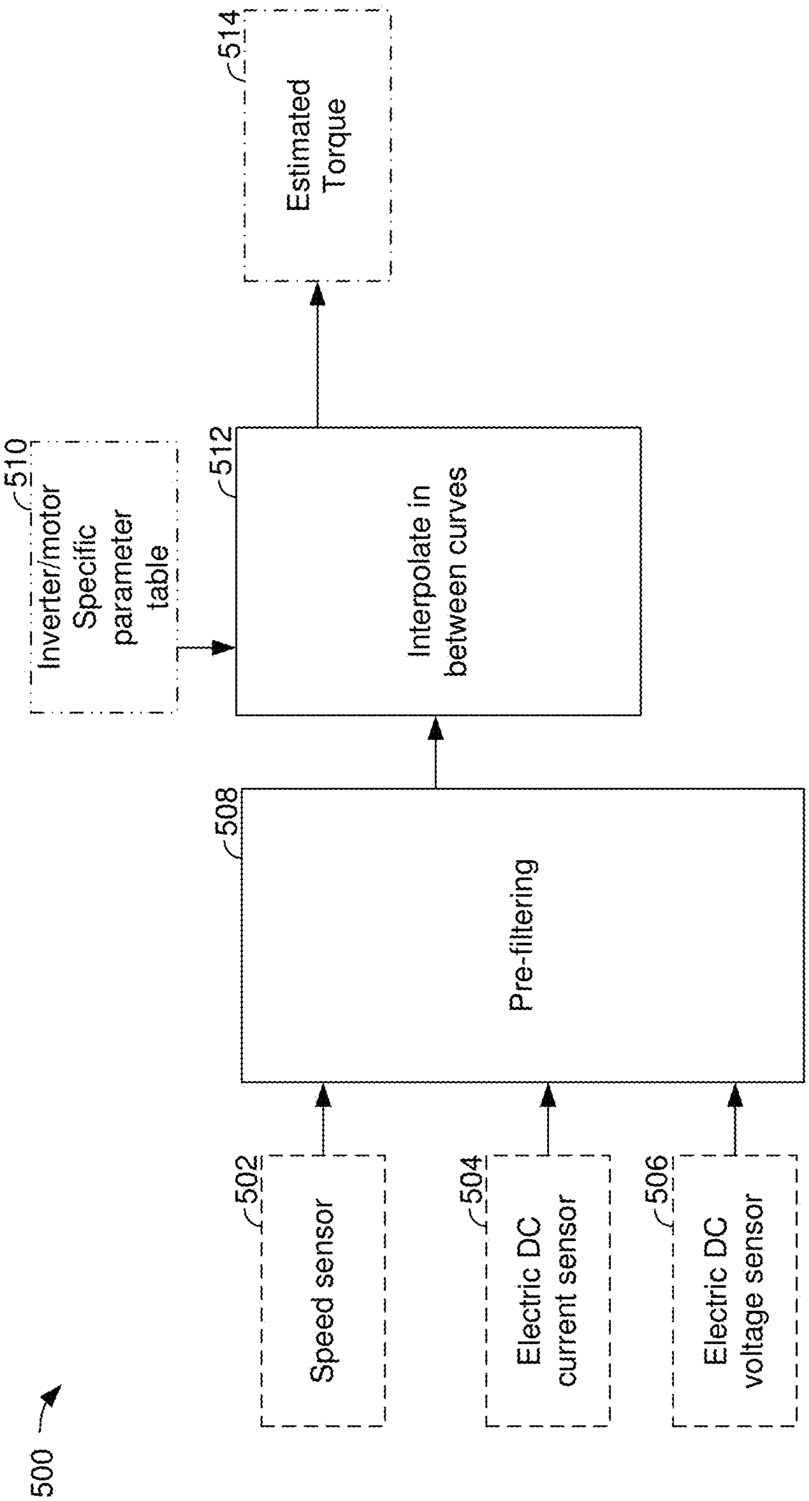

500
Speed sensor
502
Electric DC current sensor
504
Electric DC voltage sensor
506
Pre-filtering
508
Inverter/motor Specific parameter table
510
Interpolate in between curves
512
Estimated Torque
514

FIG. 5

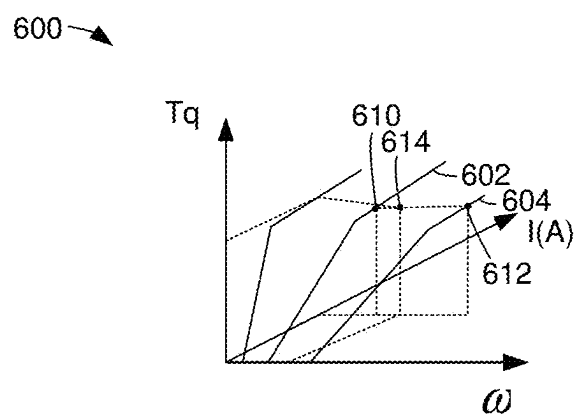


FIG. 6

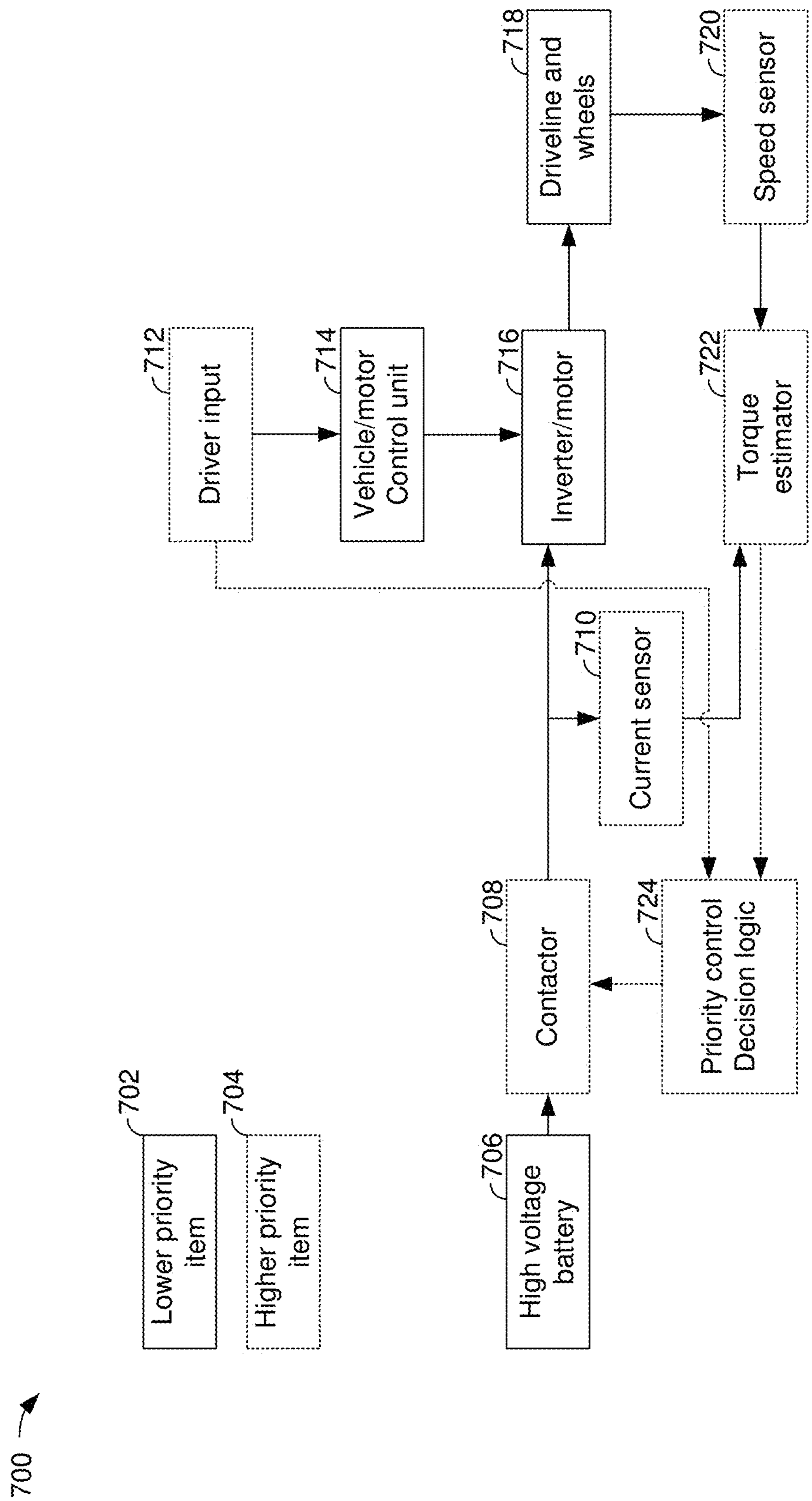

700
702
Lower priority item
704
Higher priority item
706
High voltage battery
708
Contactor
710
Current sensor
712
Driver input
714
Vehicle/motor Control unit
716
Inverter/motor
718
Driveline and wheels
720
Speed sensor
722
Torque estimator
724
Priority control Decision logic

FIG. 7

SYSTEM AND METHOD FOR ELECTRIC MACHINE TORQUE ESTIMATION

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to U.S. Provisional Application No. 63/557,955, entitled "SYSTEM AND METHOD FOR ELECTRIC MACHINE TORQUE ESTIMATION", and filed on Feb. 26, 2024. The entire contents of the above-listed application are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to a method and system for estimating torque of an electric machine. The electric machine may be a propulsion source for a vehicle.

BACKGROUND AND SUMMARY

[0003] A torque generating system for an electric or hybrid vehicle may be comprised of an inverter or power converter and an electric machine. A controller for the torque generating system may provide an estimate of an amount of torque that is being generated by the torque source. However, the calculation of the torque signal that provides the estimated amount of torque that is being generated by the torque source may not be free from interference from other calculations done in the system and it may not provide compensation for inverter losses, such that the torque signal may not be suitable for tasks that rely on higher accuracy and higher priority. Therefore, it may be desirable to provide a way to generate an independent and reliable torque estimate that is simple to generate and that may depend on fewer parameters than other estimates. Further, it may be desirable to generate a torque estimate that has a lower propensity to degrade as compared to other torque estimates.

[0004] The inventors herein have recognized the above-mentioned issues and have developed a method for estimating torque output of an electric vehicle propulsion source, comprising: generating a first polynomial for a first group of speeds of the electric vehicle propulsion source, where output of the first polynomial provides a torque estimate for the electric vehicle propulsion source; generating a second polynomial for a second group of speeds of the electric vehicle propulsion source, where output of the second polynomial provides a torque estimate for the electric vehicle propulsion source; and operating a vehicle in response to the torque estimate for the electric vehicle propulsion source.

[0005] By generating polynomials to estimate torque output of a torque generating system, it may be possible to provide the technical result of generating an electric machine torque estimate that compensates for inverter losses and reduces system complexity. In particular, a first polynomial may be generated to approximate electric machine torque at lower electric machine speeds and a second polynomial may be generated to approximate electric machine torque at higher electric machine speeds. The polynomials may simplify electric machine torque estimation and improve torque estimation reliability.

[0006] The present description may provide several advantages. In particular, the approach may provide reliable torque estimation. In addition, the approach may simplify electric machine torque estimation to reduce processing time and enable higher priority systems to rely on the torque estimated.

[0007] It is to be understood that the summary above is provided to introduce in simplified form a selection of concepts that are further described in the detailed description. It is not meant to identify key or essential features of the claimed subject matter, the scope of which is defined uniquely by the claims that follow the detailed description. Furthermore, the claimed subject matter is not restricted to implementations that solve any disadvantages noted above or in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is an illustration of an example vehicle that includes an electric machine that is configured to propel a vehicle.

[0009] FIG. **2** shows a flowchart of a method to generate curves for estimating torque output of a torque source.

[0010] FIG. **3** shows example plots representing data flow to generate curves for estimating torque output of a torque source.

[0011] FIG. **4** shows a plot of example drive and regeneration torques for an electric machine.

[0012] FIG. **5** shows a block diagram for embedded processing and data flow for estimating torque output of an electric machine.

[0013] FIG. **6** shows a plot that illustrates interpolation between two curves.

[0014] FIG. **7** shows a block diagram of integration of a method for estimating torque output of an electric machine into a vehicle.

DETAILED DESCRIPTION

[0015] A method and system for estimating torque output of an electric machine of a vehicle are described. The method and system are suitable for electric drive systems as shown in FIG. **1** and hybrid vehicles. In one example, the method and system may generate a group of curves to approximate torque output of an electric machine as shown in the block diagram of FIG. **2**. The flow of data to generate the torque estimate may be as shown in FIG. **3**. Torque output for an electric machine may be as shown in the plot of FIG. **4**. Data may be processed according to the block diagram of FIG. **5** to generate a torque estimate for an electric machine. Torque estimates may be based on interpolating between two curves as shown in FIG. **6**. A system and method for estimating torque generated via an electric machine may be integrated into a vehicle as shown by the block diagram of FIG. **7**.

[0016] FIG. **1** illustrates an example vehicle propulsion system **199** for vehicle **10**. A front end **110** of vehicle **10** is indicated and a rear end **111** of vehicle **10** is also indicated. Vehicle **10** travels in a forward direction when front end leads movement of vehicle **10**. Vehicle **10** travels in a reverse direction when rear end leads movement of vehicle **10**. Vehicle propulsion system **199** includes a propulsion source **105** (e.g., an electric machine, such as a motor), but in other examples two or more propulsion sources may be provided. In one example, propulsion source **105** may be a synchronous or induction electric machine that may operate as a motor or generator. The propulsion source **105** is fastened to the electrified axle **190** and it delivers power from its rotor

105*a* to gear set **107**. In FIG. **1** mechanical connections between the various components are illustrated as solid lines, whereas electrical connections between various components are illustrated as dashed lines.

[0017] Vehicle propulsion system **199** includes an electrified axle **190** (e.g., an axle that includes an integrated electric machine that provides propulsive effort for the vehicle). Electrified axle **190** may include two axle shafts, including a first or right axle shaft **190***a* and a second or left axle shaft **190***b*. Vehicle **10** further includes front wheels **102** and rear wheels **103**.

[0018] The electrified axle **190** may be an integrated axle that includes differential gears **106**, gear set **107**, and propulsion source **105**. Gear set **107** may be referred to as a step ratio transmission and it may include a plurality of gears and one or more clutches to shift between gears. In one example, the gear set may include a first gear **108**, a second gear **109**, and a third gear **131**. Further, the electrified axle **190** may include one or more clutch actuators **112** to shift one or more clutches **145**. Clutches **145** may be configured as dog clutches, wet, friction, curvic, or other known configuration. Encoder **168** provides an indication of motor position and a position of a half of clutch **145** (e.g., upstream power path side of clutch **145**) and encoder **169** provides an indication of a clutch position of a downstream clutch power path side of clutch **145**. Electrified axle **190** may include a first speed sensor **119** for sensing a speed of propulsion source **105** on an input side of clutch **145** and a second speed sensor **122** for sensing a speed of an output side of clutch **145**. Electric power inverter **115** is electrically coupled to propulsion source **105**. An axle control unit **116** is electrically coupled to sensors and actuators of electrified axle **190**.

[0019] Propulsion source **105** may transfer mechanical power to or receive mechanical power from gear set **107**. As such, gear set **107** may be a multi-speed gear set that may shift between gears (e.g., first gear **108** and second gear **109**) when commanded via axle control unit **116**. Axle control unit **116** includes a processor **116***a* and memory **116***b*. Memory **116***b* may include read only memory, random access memory, and keep alive memory. Axle control unit **116** may receive transmission temperature via sensor **133**. Gear set **107** may transfer mechanical power to or receive mechanical power from differential gears **106**. Differential gears **106** may transfer mechanical power to or receive mechanical power from rear wheels **103** via right axle shaft **190***a* and left axle shaft **190***b*. Propulsion source **105** may consume alternating current (AC) electrical power provided via electric power inverter **115**. Alternatively, propulsion source **105** may provide AC electrical power to electric power inverter **115**. Electric power inverter **115** may be provided with high voltage direct current (DC) power from battery **160** (e.g., a traction battery, which also may be referred to as an electric energy storage device or battery pack). Electric power inverter **115** may convert the DC electrical power from battery **160** into AC electrical power for propulsion source **105**. Alternatively, electric power inverter **115** may be provided with AC power from propulsion source **105**. Electric power inverter **115** may convert the AC electrical power from propulsion source **105** into DC power to store in battery **160**.

[0020] Battery **160** may periodically receive electrical energy from a power source such as a stationary power grid **5** residing external to the vehicle (e.g., not part of the vehicle). As a non-restricted example, vehicle propulsion system **199** may be configured as a plug-in electric vehicle (EV), whereby electrical energy may be supplied to battery **160** via the power grid **5** and charging station **12**. Electric charge may be delivered to battery **160** via plug receptacle **100**.

[0021] Battery **160** may include a BMS controller **139** (e.g., a battery management system controller) and an electrical power distribution box **162**. BMS controller **139** may provide charge balancing between energy storage elements (e.g., battery cells) and communication with other vehicle controllers (e.g., vehicle control unit **152**). BMS controller **139** includes a core processor **139***a* and memory **139***b* (e.g., random-access memory, read-only memory, and keep-alive memory).

[0022] Vehicle **10** may include a vehicle control unit (VCU) **152** that may communicate with electric power inverter **115**, axle control unit **116**, friction or foundation caliper controller **170**, global positioning system (GPS) **188**, BMS controller **139**, and dashboard **130** and components included therein via controller area network (CAN) **120**. VCU **152** includes memory **114**, which may include read-only memory (ROM or non-transitory memory) and random access memory (RAM). VCU also includes a digital processor or central processing unit (CPU) **153**, and inputs and outputs (I/O) **118** (e.g., digital inputs including counters, timers, and discrete inputs, digital outputs, analog inputs, and analog outputs). VCU may receive signals from sensors **154** and provide control signal outputs to actuators **156**. Sensors **154** may include but are not restricted to lateral accelerometers, longitudinal accelerometers, yaw rate sensors, inclinometers, temperature sensors, battery voltage and current sensors, and other sensors described herein. Additionally, sensors **154** may include steering angle sensor **197**, driver demand pedal position sensor **141**, vehicle range finding sensors including radio detection and ranging (RADAR), light detection and ranging (LIDAR), sound navigation and ranging (SONAR), and caliper pedal position sensor **151**. Actuators may include but are not constrained to inverters, transmission controllers, display devices, human/machine interfaces, friction braking systems, and battery controller described herein.

[0023] Driver demand pedal position sensor **141** is shown coupled to driver demand pedal **140** for determining a degree of application of driver demand pedal **140** by human **142**. Caliper pedal position sensor **151** is shown coupled to caliper pedal **150** for determining a degree of application of caliper pedal **150** by human **142**. Steering angle sensor **197** is configured to determine a steering angle according to a position of steering wheel **198**.

[0024] Vehicle propulsion system **199** is shown with a global position determining system **188** that receives timing and position data from one or more GPS satellites **189**. Global positioning system may also include geographical maps in ROM for determining the position of vehicle **10** and features of roads that vehicle **10** may travel on.

[0025] Vehicle propulsion system **199** may also include a dashboard **130** that an operator of the vehicle may interact with. Dashboard **130** may include a display system **132** configured to display information to the vehicle operator. Display system **132** may comprise, as a non-restricting example, a touchscreen, or human machine interface (HMI), display which enables the vehicle operator to view graphical information as well as input commands. In some examples, display system **132** may be connected wirelessly to the internet (not shown) via VCU **152**. As such, in some examples, the vehicle operator may communicate via display system **132** with an internet site or software application (app) and VCU **152**.

[0026] Dashboard **130** may further include an operator interface **136** via which the vehicle operator may adjust the operating status of the vehicle. Specifically, the operator interface **136** may be configured to activate and/or deactivate operation of the vehicle driveline (e.g., propulsion source **105**) based on an operator input. Further, an operator may request an axle mode (e.g., park, reverse, neutral, drive) via the operator interface. Various examples of the operator interface **136** may include interfaces that utilize a physical apparatus, such as a key, that may be inserted into the operator interface **136** to activate the electrified axle **190** and propulsion source **105** and to turn on the vehicle **10** or may be removed to shut down the electrified axle and propulsion source **105** to turn off vehicle **10**. Electrified axle **190** and propulsion source **105** may be activated via supplying electric power to propulsion source **105** and/or electric power inverter **115**. Electrified axle **190** and electric machine may be deactivated by ceasing to supply electric power to electrified axle **190** and propulsion source **105** and/or electric power inverter **115**. Still other examples may additionally or optionally use a start/stop button that is manually pressed by the operator to start or shut down the electrified axle **190** and propulsion source **105** to turn the vehicle on or off. In other examples, a remote electrified axle or electric machine start may be initiated remote computing device (not shown), for example a cellular telephone, or smartphone-based system where a user's cellular telephone sends data to a server and the server communicates with the vehicle control unit **152** to activate the electrified axle **190** including an inverter and electric machine. Spatial orientation of vehicle **10** is indicated via axes **175**.

[0027] Vehicle **10** is also shown with a foundation or friction caliper controller **170**. Friction caliper controller **170** may selectively apply and release friction calipers (e.g., **172***a* and **172***b*) via allowing hydraulic fluid to flow to the friction calipers. The friction calipers may be applied and released so as to reduce a possibility of locking of the friction calipers to front wheels **102** and rear wheels **103**. Wheel position or speed sensors **161** may provide wheel speed data to friction caliper controller **170**. Vehicle propulsion system **199** may provide torque to rear wheels **103** to propel vehicle **10**.

[0028] A human or autonomous driver may request a driver demand wheel torque, or alternatively a driver demand wheel power, via applying driver demand pedal **140** or via supplying a driver demand wheel torque/power request to vehicle control unit **152**. Vehicle control unit **152** may then demand a torque or power from propulsion source **105** via commanding axle control unit **116**. Axle control unit **116** may command electric power inverter **115** to deliver the driver demand wheel torque/power via electrified axle **190** and propulsion source **105**. Electric power inverter **115** may convert DC electrical power from battery **160** into AC power and supply the AC power to propulsion source **105**. Propulsion source **105** rotates and transfers torque/power to gear set **107**. Gear set **107** may supply torque from propulsion source **105** to differential gears **106**, and differential gears **106** transfer torque from propulsion source **105** to rear wheels **103** via axle shafts **190***a* and **190***b*.

[0029] During conditions when the driver demand pedal is fully released, vehicle control unit **152** may request a small negative or regenerative braking power to gradually slow vehicle **10** when a speed of vehicle **10** is greater than a threshold speed. The amount of regenerative braking power requested may be a function of driver demand pedal position, battery state of charge (SOC), vehicle speed, and other conditions. If the driver demand pedal **140** is fully released and vehicle speed is less than a threshold speed, vehicle control unit **152** may request a small amount of positive torque/power (e.g., propulsion torque) from propulsion source **105**, which may be referred to as creep torque or power. The creep torque or power may allow vehicle **10** to remain stationary when vehicle **10** is on a small positive grade.

[0030] The human or autonomous driver may also request a negative or regenerative driver demand braking torque, or alternatively a driver demand braking power, via applying caliper pedal **150** or via supplying a driver demand braking power request to vehicle control unit **152**. Vehicle control unit **152** may request that a first portion of the driver demanded braking power be generated via electrified axle **190** and propulsion source **105** via commanding axle control unit **116**. Additionally, vehicle control unit **152** may request that a portion of the driver demanded braking power be provided via friction calipers **172** via commanding friction caliper controller **170** to provide a second portion of the driver requested braking power.

[0031] After vehicle control unit **152** determines the braking power request, vehicle control unit **152** may command axle control unit **116** to deliver the portion of the driver demand braking power allocated to electrified axle **190**. Electric power inverter **115** may convert AC electrical power generated by propulsion source **105** into DC power for storage in battery **160**. Propulsion source **105** may convert the vehicle's kinetic energy into AC power.

[0032] Axle control unit **116** includes predetermined transmission gear shift schedules whereby fixed ratio gears of gear set **107** may be selectively engaged and disengaged. Shift schedules stored in axle control unit **116** may select gear shift points or conditions as a function of driver demand wheel torque and vehicle speed.

[0033] The description herein is not limited to electric vehicles configured as shown in FIG. **1**. Rather, the description is applicable to other electrified powertrains including parallel and series hybrid vehicles.

[0034] Thus, the system of FIG. **1** provides for a vehicle system, comprising: an electric energy storage device; an electric machine configured to propel a vehicle; an inverter configured to supply alternating current to the electric machine; a contactor arranged in series between the electric energy storage device and the inverter; and one or more controllers (e.g., controllers **152** and/or **116** of FIG. **1**) including executable instructions stored in non-transitory memory that cause the controller to selectively open the contactor in response to an estimate of torque output of the electric machine, where the estimate of torque output is based on interpolating between two linear curves or interpolating between output of two polynomials. In a first example, the vehicle system includes where a first polynomial of the two polynomials corresponds to a first speed range of the electric machine. In a second example that may include the first example, the vehicle system includes where a second polynomial of the two polynomials corresponds to a second speed range of the electric machine, the second speed range greater than the first speed range. In a third example that may include one or both of the first and second examples, the vehicle system includes where the first polynomial receives electric machine speed and inverter input current as inputs. In a fourth example that may include one or more of the first through third examples, the vehicle system includes where a first curve of the two linear curves is based on a DC current flow into the inverter. In a fifth example that may include one or more of the first through fourth examples, the vehicle system includes where a second curve of the two linear curves is based on a take-over point. In a sixth example that may include one or more of the first through fifth examples, the vehicle system includes where the second curve is further based on DC current flow into the inverter.

[0035] Turning now to FIG. **2**, a block diagram of a method **200** that describes determination of parameters for a torque estimation algorithm is shown. The method of FIG. **2** may operate in cooperation with the systems and other methods described herein. Further, the method of FIG. **2** may be incorporated as executable instructions that are stored in non-transitory memory of one or more controllers (e.g., controllers **152** and/or **116** of FIG. **1**) that cause the controller to adjust states or positions of actuators in the physical world. The blocks (e.g., **204**, **208**, and **212**) that are formed by dashed lines represent processing blocks where data may be processed and blocks that are formed by solid lines (e.g., **202**, **206**, **210**, and **214**) represent passive data blocks.

[0036] At **202**, method **200** generates an inverter and motor loss table. In one example, the inverter and motor loss table is a two dimensional table that may be referenced by DC current that is input to the inverter and motor and motor speed. The table may have dimensions of N×M where N is the number of motor speed rows in the table and M is the number of motor torque command columns. Thus, the table includes N×M speed and torque combinations. Loss values in the table may be determined by subtracting measured motor power output from electric power that is input to the inverter and motor. Alternatively, a percentage loss value may be determined by dividing the motor output power by the inverter and motor input electric power. The losses determined at **202** are input to block **204**.

[0037] At block **204**, method **200** estimates an effective torque that is output by the motor. In one example, method **200** may estimate the effective torque from a second table that may be referenced by DC current that is input to the inverter and motor and motor speed. The second table may also have dimensions of N rows by M columns. The second table outputs an effective torque value and the effective torque value may include compensation for the loss value that was generated at **202**. In particular, the effective torque generated by the electric machine may be determined via the equation: Trq_EFF=(V_DC*I_DC-losses)/ω, where Trq_EFF is the effective torque, V_DC is the input DC voltage to the inverter, I_DC is the input current to the inverter, losses are the inverter and electric machine losses, and ω is the rotational speed of the electric machine. The losses may be determined by an amount of power to yield a particular effective torque and may be determined via the following equations: input power=V_DC*I_DC=losses+mechanical power generated by the electric machine=losses+(ω*Trq_eff). The effective torque values at predetermined electric machine rotational speeds and predetermined DC current values that are input to the inverter that supplies power to the electric machine may be inserted into a data structure (e.g., a table or function) as shown in FIG. **3**.

[0038] The effective torque value at present operating conditions may be estimated by converting the table mentioned at **202**, which outputs the required DC current for particular electric machine speed and torque combinations into a table that outputs effective electric machine torque for particular electric machine speed and DC input current (e.g., DC current into the electric machine's associated inverter) combinations into a table that outputs effective electric machine torque according to electric machine speed and DC input current combinations. Method **200** proceeds to **206**.

[0039] At **206**, method **200** decides whether to process the table that outputs the effective torque value at block **204**. The table may be processed via a first method or a second method. If the first method is selected, method **200** proceeds to **208**. If the second method is selected, method **200** proceeds to **212**. The first method may be selected based on whether or not the processor or controller has capacity to execute the polynomial calculation in real-time. In order to provide consistent and repeatable operation of devices and sub-systems, a controller may begin execution of certain controller executable instructions at fixed time intervals between the beginnings of executing the certain controller executable instructions (e.g., real-time time intervals). If the instructions to perform the first method are executable within a particular real-time interval/window and/or if the controller includes a floating point library or processor, method **200** may select the first method to execute. The second method may be chosen when a floating point library or processor are not present.

[0040] At **208**, method **200** derives an approximation for the table of step **204** according to a best fit algorithm. The best fit algorithm minimizes both the total amount inaccuracy as well as the maximum level of inaccuracy in a configurable weighted manner. Method **200** approximates the actual torque via two polynomials, where actual torque=polynomial_low_speed=f_low (speed, current)=$\Sigma_{i,j=0}^{n} a_{ij} x^i y^j$ for $\text{speed} < \text{low\_speed\_threshold}$, where actual torque=polynomial_high_speed=f_high(speed, current)=$\Sigma_{i,j=0}^{n} a_{ij} x^i y^j + b*(y/x)$ for $\text{speed} > \text{low\_speed\_threshold}$. The variable speed is the motor speed and the variable current is DC current flowing to the inverter that supplies power to the motor. Further, x=speed, y=current, such that a set of a_{ij} and b parameters are constants of the polynomial that describes a three dimensional surface between electric machine effective torque, electric machine speed, and current flow inout the inverter that provides power to the electric machine. Practically, i=2 and y=4 gives a good fit for the low speed region. Additionally, i=3 and y=2 gives a good fit for the high speed region. The size of the required data set for the embedded processing is 2×i×j. The polynomials Polynomial_low_speed and Polynomial_high_speed are stored in controller memory at block **210** and the actual motor torque may be determined from the polynomials.

[0041] Additionally, method **200** may apply a motor data set that includes a maximum DC voltage input to the inverter so that applied torque may not be under estimated. Alternatively, the motor data set of an estimated average input DC voltage may be applied to generate an estimated torque that is on average more accurate for a full range of DC input voltages. In still other examples, different curves may be obtained for different DC voltage motor data to increase accuracy of motor torque estimates. Further, different data sets may be applied for estimating motoring and regeneration electric machine torques, where different polynomials or linear approximations may be provided for motoring and regeneration torques.

[0042] At **212**, method **200** derives an approximation for the table of step **204** according to a second algorithm. The second method derives a reasonable approximation for the above table using a best fit algorithm. The second algorithm minimizes both the total amount inaccuracy of motor torque as well as the maximum level of inaccuracy of motor torque in a configurable weighted manner. The actual motor torque is approximated at a given motor speed by two linear approximations, where actual motor torque=a_1*motor current for motor current<take over point, actual motor torque=a_1*take-over point+a_2*(motor current-take over point) for motor current>take over point for a well-chosen set of motor speeds a_1, a_2, and take over point will be determined. The take-over point is a given value of DC current below which the torque increases with increasing current by a factor of a1; above which the torque increases with increasing current by a factor of a2. The variables a1 and a2 are defined as the factor of increase of torque versus an increase of current. For example, a1=5 Newton-meter (Nm)/ampere so an increase of 1 Amp DC current, leads to an increase of 5 Nm of torque. All three values a1, a2, and take-over point are obtained by applying a best fit algorithm to approach the surface Trq=f (I_DC, speed) for a particular electric machine speed. The size of the obtained data set for the embedded processing is typically 3×N, where N is the amount of motor speed set points, and where 3 is given by two slopes of the approximation curves and one take over point.

[0043] At **214**, method **200** a best fit of the surface Trq=f (I_DC, speed) formed by a set of lines for different electric machine speeds is determined. All three values a1, a2, and take-over point are obtained by applying a best fit algorithm to approach the surface Trq=f (I_DC, speed) given a predetermined electric machine speed. The best fit is performed for the entire speed range so that an electric machine speed of 100 revolutions/minute (RPM) provides a triple (a1, a2, take-over point), an electric machine speed of 200 RPM provides another triple (a1, a2, take over point), etc.

[0044] Referring now to FIG. **3**, plots illustrating flow of data to determine motor torque approximation curves are shown. The left most plot in FIG. **3** is a plot of motor data points and corresponding inverter and motor losses for the motor torque and motor rotational speed set points. The middle plot in FIG. **3** is a plot of data that may be stored in a table that may be referenced by motor rotational speed and DC (direct current) that is supplied to the inverter that supplies electric power to the motor (I(A)). The table is populated with motor torque estimates and outputs the motor torque estimates according to present motor rotational speed and DC current flow into the inverter. The right most plot in FIG. **3** is a plot showing curves that describe motor torque (Trq) as a function of motor rotational speed (ω) and DC current flow into the inverter (I(A)). The curves in the right most plot may be generated from the data in the table that is the basis for the center plot in FIG. **3**. Plots similar to the plots in FIG. **3** may be generated for regeneration torque and motoring torque.

[0045] Referring now to FIG. **4**, a plot that shows drive or motoring torque and regenerative torque versus electric machine current for a predetermined electric machine rotational speed are shown. The vertical axis represents electric machine torque and the horizontal axis represents electric machine current. Curve **404** represents motoring torque for an electric machine versus electric current for the electric machine. Curve **402** represents regeneration torque for an electric machine versus electric current for the electric machine. Since these curves are different, they may be approximated via different polynomials or different linear approximations.

[0046] Moving on to FIG. **5**, a block diagram of a method **500** for embedded processing of electric machine data is shown. The method of FIG. **5** may operate in cooperation with the systems and other methods described herein. Further, the method of FIG. **5** may be incorporated as executable instructions that are stored in non-transitory memory of a controller that cause the controller to adjust states or positions of actuators in the physical world. The blocks that are formed by dashed lines (e.g., **502**, **504**, and **506**) represent blocks where signals may be captured via a controller storing data values to controller memory. The blocks that are formed by solid lines (e.g., **508** and **512**) represent data processing blocks. The block that is formed by dash-dot-dot a line (e.g., **510**) represents a data table. The block that is formed by a dash-dot line (e.g., **514**) represents an output block where data is output.

[0047] Method **500** includes an electric machine rotational speed sensor that is indicated at block **502**. The rotational speed sensor may measure a speed of an electric machine rotor or a device that is coupled to the electric machine's rotor. A speed signal is generated by the speed sensor and the speed signal is provided to a pre-filtering block **508**. Method **500** also includes a DC current sensor and the DC current sensor senses DC current that flows into and out of the inverter that is electrically coupled to the electric machine or propulsion source. An electric current signal is generated by the current sensor and the current signal is provided to the pre-filtering block **508**. Method **500** also includes a DC voltage sensor and the DC voltage sensor senses a voltage that is input and output of the inverter that is electrically coupled to the electric machine. A voltage signal is generated by the voltage sensor and the voltage signal is input to the pre-filtering block **508**.

[0048] Pre-filtering block **508** filters the electric machine speed, DC current, and DC voltage signals. In one example, method **500** may low pass filter the signals to reduce noise that may be included in the signals. The filtered electric machine speed, DC current, and DC voltage are provided to block **512** where method **500** interpolates between curves that describe a relationship between electric machine speed, DC current flow into or out of an inverter or power converter and the electric machine that provides propulsive effort to the vehicle, and torque output of the electric machine. The interpolation may be performed via processing block **522** applying values stored in data table **520** that have been determined via method **200**. The values may represent polynomials or linear approximations of curves that describe electric machine torque output according to DC current and electric machine rotational speed as discussed with respect to method **200**. The interpolation may be performed via a third method or a fourth method. The third method includes selecting an applicable polynomial based on electric machine rotational speed, applying the selected polynomial by plugging (e.g., inserting) the present electric machine speed and DC current flow into or out of the inverter that is coupled to the electric machine into the polynomial and generating the electric machine torque according to output of the selected polynomial. Optionally, the DC voltage that is input or output of the inverter may be plugged into the polynomial when the polynomial is based on DC voltage. On the other hand, the fourth method includes selecting two linear interpolation data sets that are closest to the present rotational speed of the electric machine, determine the electric machine torque according to each of the interpolation data sets according to the DC current flow into or out of the inverter, and interpolate between the two torques determined from the two interpolation data sets. Optionally, if DC voltage into or out of the inverter that is electrically coupled to the electric machine is available, different interpolation curves may be applied to estimate electric machine torque according to the DC voltage.

[0049] The estimated electric machine torque may be supplied output block **514** where the estimated electric machine torque may be made available throughout the system for use as a controller feedback signal or control signal. The estimated electric machine torque may be made available to electric machine control routines as indicated at block **514**.

[0050] Referring now to FIG. **6**, a plot showing an example of interpolation between two different curves to estimate torque output of an electric machine is shown. Plot **600** shows a first relationship between DC current (I(A)), electric machine rotational speed (ω), and electric machine torque (Trq) that is represented by curve **602**. Plot **600** shows a second relationship between DC current (I(A)), electric machine rotational speed (ω), and electric machine torque (Trq) that is represented by curve **604**. For estimating a torque of the electric machine that is generated at an electric machine speed that is in between the speed of curve **602** and curve **604**, a method interpolates between a torque at the present DC current for curve **602** and torque at the present DC current for curve **604**. In particular, the method subtracts the present speed of the electric machine from the electric machine speed represented by curve **602** and divides the result by the speed represented by curve **604** minus the speed represented by curve **602**. The result of the division is multiplied by the torque represented by curve **604** (e.g., torque at dot **610**) minus the torque represented by curve **602** (e.g., torque at dot **612**). The result of the multiplication is added to the torque that is represented by curve **604** at the present DC current and electric machine speed to generate the interpolated torque value (e.g., torque at dot **614**).

[0051] Finally, FIG. **7** shows a block diagram **700** of torque estimator that has been integrated into a vehicle. The methods of FIGS. **2** and **5** are represented by block **722** that is labeled "torque estimator." The blocks that are shown in dashed lines may be hosted in a high priority controller (e.g., controllers **152** and/or **116** of FIG. **1**). The high priority controller may be specifically rated to handle higher priority system tasks and calculations that may include but are not limited to automatic system shutdown and processing higher priority inputs.

[0052] Block diagram may include lower priority tasks and inputs/outputs that are represented by block **702** and higher priority tasks and inputs/outputs that are represented by block **704**. A high voltage battery represented by block **706** may supply electric power and the supply of electric power may be controlled via selectively opening and closing contactor **708**. Contactor **708** may be selectively opened and closed according to priority control decision logic that is represented by block **724**. Electric current flow through contactor **708** may be monitored via current sensor **710** and supplied to the inverter and motor that is represented by block **716** (e.g., inverter **115** and electric machine **105** of FIG. **1**). Human driver input that is represented by block **712** may be provided to priority control decision logic block **724** and a vehicle controller that is represented by block **714**. The electric machine may provide propulsive effort to the vehicle driveline and wheels that are represented by block **718**. Wheel speed and/or electric motor rotational speed may be sensed via a speed sensor that is represented by block **720** and the speed may be input to torque estimator block **722**.

[0053] In one example, the priority control decision logic may open contactor **708** in response to torque estimates that are generated according to torque estimator **722**. Further, output from torque estimator **722** may be applied in closed loop speed and/or torque control routines to control torque of the electric machine in block **716**. For example, if a drive demands 200 Newton-meters (Nm) output torque from the inverter and motor at block **716**, but torque estimator indicates that torque output of the electric machine is 202 Nm, electric current supplied to the electric machine of block **716** may be reduced to reduce estimated electric machine torque to 200 Nm.

[0054] Thus, the methods described herein provide for a method for estimating torque output of an electric vehicle propulsion source, comprising: generating a first polynomial for a first group of speeds of the electric vehicle propulsion source, where output of the first polynomial provides a torque estimate for the electric vehicle propulsion source; generating a second polynomial for a second group of speeds of the electric vehicle propulsion source, where output of the second polynomial provides a torque estimate for the electric vehicle propulsion source; and operating a vehicle in response to the torque estimate for the electric vehicle propulsion source. In a first example, the method includes where the first polynomial corresponds to a first speed range of the electric vehicle propulsion source. In a second example that may include the first example, the method includes where the second polynomial corresponds to a second speed range of the electric vehicle propulsion source, the second speed range greater than the first speed range. In a third example that may include one or both of the first and second examples, the method includes where the first polynomial receives electric machine speed and inverter DC input current as inputs. In a fourth example that may include one or more of the first through third examples, the method includes where the second polynomial receives electric machine speed and inverter DC input current as inputs. In a fifth example that may include one or more of the first through fourth examples, the method includes where the torque estimate includes compensation for inverter losses. In a sixth example that may include one or more of the first through fifth examples, the method includes where operating the vehicle includes adjusting torque output of the electric vehicle propulsion source. In a seventh example that may include one or more of the first through sixth examples, the method includes where operating the vehicle includes opening a contactor that selectively supplies electric power to the electric vehicle propulsion source.

[0055] The methods also provides for a method for estimating torque output of an electric vehicle propulsion source, comprising: generating a first linear curve for a first group of speeds of the electric vehicle propulsion source, where output of the first linear curve provides a torque estimate for the electric vehicle propulsion source; generating a second linear curve for a second group of speeds of the electric vehicle propulsion source, where output of the second linear curve is a torque estimate for the electric vehicle propulsion source; and operating a vehicle in response to the torque estimate for the electric vehicle propulsion source. In a first example, the method includes where the first linear curve is based on DC current flow into an inverter. In a second example that may include the first example, the method includes where the second linear curve is based on DC current flow into an inverter. In a third example that may include one or both of the first and second examples, the method includes where the second linear curve is further based on take-over point. In a fourth example that may include one or both of the first through third examples, the method includes where the second linear curve is further based on a speed of the electric vehicle propulsion source.

[0056] Note that the example control and estimation routines included herein can be used with various powertrain and/or vehicle system configurations. The control methods and routines disclosed herein may be stored as executable instructions in non-transitory memory and may be carried out by the control system including the controller in combination with the various sensors, actuators, and other transmission and/or vehicle hardware. Further, portions of the methods may be physical actions taken in the real world to change a state of a device. Thus, the described actions, operations and/or functions may graphically represent code to be programmed into non-transitory memory of the computer readable storage medium in the vehicle and/or transmission control system. The specific routines described herein may represent one or more of any number of processing strategies such as event-driven, interrupt-driven, multi-tasking, multi-threading, and the like. As such, various actions, operations, and/or functions illustrated may be performed in the sequence illustrated, in parallel, or in some cases omitted. Likewise, the order of processing is not necessarily required to achieve the features and advantages of the examples described herein, but is provided for ease of illustration and description. One or more of the illustrated actions, operations and/or functions may be repeatedly performed depending on the particular strategy being used. One or more of the method steps described herein may be omitted if desired.

[0057] While various embodiments have been described above, it is to be understood that they have been presented by way of example, and not limitation. It will be apparent to persons skilled in the relevant arts that the disclosed subject matter may be embodied in other specific forms without departing from the spirit of the subject matter. The embodiments described above are therefore to be considered in all respects as illustrative, not restrictive. As such, the configurations and routines disclosed herein are exemplary in nature, and that these specific examples are not to be considered in a limiting sense, because numerous variations are possible. For example, the above technology can be applied to electric vehicles and hybrid vehicles including induction and synchronous electric machines. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed herein.

[0058] The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. These claims may refer to "an" element or "a first" element or the equivalent thereof. Such claims may be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Other combinations and sub-combinations of the disclosed features, functions, elements, and/or properties may be claimed through amendment of the present claims or through presentation of new claims in this or a related application. Such claims, whether broader, narrower, equal, or different in scope to the original claims, also are regarded as included within the subject matter of the present disclosure.

1. A method for estimating torque output of an electric vehicle propulsion source, comprising:

generating a first polynomial for a first group of speeds of the electric vehicle propulsion source, where output of the first polynomial provides a torque estimate for the electric vehicle propulsion source;

generating a second polynomial for a second group of speeds of the electric vehicle propulsion source, where output of the second polynomial provides the torque estimate for the electric vehicle propulsion source; and

operating a vehicle in response to the torque estimate for the electric vehicle propulsion source.

2. The method of claim **1**, where the first polynomial corresponds to a first speed range of the electric vehicle propulsion source.

3. The method of claim **2**, where the second polynomial corresponds to a second speed range of the electric vehicle propulsion source, the second speed range greater than the first speed range.

4. The method of claim **1**, where the first polynomial receives electric machine speed and inverter DC input current as inputs.

5. The method of claim **1**, where the second polynomial receives electric machine speed and inverter DC input current as inputs.

6. The method of claim **5**, where the torque estimate includes compensation for inverter losses.

7. The method of claim **1**, where operating the vehicle includes adjusting torque output of the electric vehicle propulsion source.

8. The method of claim **1**, where operating the vehicle includes opening a contactor that selectively supplies electric power to the electric vehicle propulsion source.

9. A vehicle system, comprising:

an electric energy storage device;

an electric machine configured to propel a vehicle;

an inverter configured to supply alternating current to the electric machine;

a contactor arranged in series between the electric energy storage device and the inverter; and

a controller including executable instructions stored in non-transitory memory that cause the controller to selectively open the contactor in response to an estimate of torque output of the electric machine, where the estimate of torque output is based on interpolating between two linear curves or interpolating between output of two polynomials.

10. The vehicle system of claim **9**, where a first polynomial of the two polynomials corresponds to a first speed range of the electric machine.

11. The vehicle system of claim **10**, where a second polynomial of the two polynomials corresponds to a second speed range of the electric machine, the second speed range greater than the first speed range.

12. The vehicle system of claim **11**, where the first polynomial receives electric machine speed and inverter input current as inputs.

13. The vehicle system of claim **9**, where a first curve of the two linear curves is based on a DC current flow into the inverter.

14. The vehicle system of claim **13**, where a second curve of the two linear curves is based on a take-over point.

15. The vehicle system of claim **14**, where the second curve is further based on DC current flow into the inverter.

16. A method for estimating torque output of an electric vehicle propulsion source, comprising:

generating a first linear curve for a first group of speeds of the electric vehicle propulsion source, where output of the first linear curve provides a torque estimate for the electric vehicle propulsion source;

generating a second linear curve for a second group of speeds of the electric vehicle propulsion source, where output of the second linear curve is the torque estimate for the electric vehicle propulsion source; and

operating a vehicle in response to the torque estimate for the electric vehicle propulsion source.

17. The method of claim **16**, where the first linear curve is based on DC current flow into an inverter.

18. The method of claim **17**, where the second linear curve is based on DC current flow into the inverter.

19. The method of claim **18**, where the second linear curve is further based on take-over point.

20. The method of claim **16**, where the second linear curve is further based on a speed of the electric vehicle propulsion source.

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